aula - Navier-Stokes CFD

aanascimento

May 2026

1 Discretização

$$\frac{\partial u}{\partial t} + \left(\frac{\partial uu}{\partial x} + \frac{\partial uv}{\partial y} + \frac{\partial uw}{\partial z} \right) = -\frac{1}{\rho}\frac{\partial P}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \tag{1}$$

fazer a integral no volume infinitezimal e no tempo

$$\int \frac{\partial u}{\partial t} \cdot dt \cdot dVol = (u^t - u^0) \cdot \Delta Vol \tag{2}$$

$$\begin{split} \int \left(\frac{\partial uu}{\partial x} + \frac{\partial uv}{\partial y} \right) \cdot dt \cdot dVol = (u_e u_e - u_w u_w) \cdot \Delta y \cdot \Delta z \Delta t + \\ (u_n v_n - u_s v_s) \cdot \Delta x \cdot \Delta z \Delta t + (u_t w_t - u_b w_b) \cdot \Delta x \cdot \Delta y \Delta t \end{split} \tag{3}$$

$$\int \frac{1}{\rho}\frac{\partial P}{\partial x} \cdot dt \cdot dVol = \frac{1}{\rho}(P_e - P_w) \cdot \Delta y \cdot \Delta z \Delta t \tag{4}$$

$$\begin{split} \int \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \cdot dt \cdot dVol = \nu \Bigg(\frac{u_E - 2u_P + u_W}{\Delta x} \cdot \Delta y \cdot \Delta z \Delta t \\ + \frac{u_N - 2u_P + u_S}{\Delta y} \cdot \Delta x \cdot \Delta z \Delta t \\ + \frac{u_T - 2u_P + u_B}{\Delta z} \cdot \Delta y \cdot \Delta x \Delta t \Bigg) \end{split} \tag{5}$$

- escoamento incompressível ($\nabla . \vec{V} = 0$)
- fluido newtoniano ($\nu = cte$)
- sem efeitos de turbulencia = 2D
- propriedades constantes = $\rho = cte$
- sem termo fonte
- método explicito

Fazendo a divisão pelo volume infinitezimal e pelo tempo

$$\frac{u^t - u^0}{\Delta t} + \frac{u_e^0 u_e^0 - u_w^0 u_w^0}{\Delta x} + \frac{u_n^0 v_n^0 - u_s^0 v_s^0}{\Delta y} = -\frac{1}{\rho}\frac{P_e^0 - P_w^0}{\Delta x} + \nu\left(\frac{u_E^0 - 2u_P^0 + u_W^0}{\Delta x^2} + \frac{u_N^0 - 2u_P^0 + u_S^0}{\Delta y^2}\right) \tag{6}$$

Direção y

$$\frac{v^t - v^0}{\Delta t} + \frac{u_e^0 v_e^0 - u_w^0 v_w^0}{\Delta x} + \frac{v_n^0 v_n^0 - v_s^0 v_s^0}{\Delta y} = -\frac{1}{\rho}\frac{P_n^0 - P_s^0}{\Delta y} + \nu\left(\frac{v_E^0 - 2v_P^0 + v_W^0}{\Delta x^2} + \frac{v_N^0 - 2v_P^0 + v_S^0}{\Delta y^2}\right) \tag{7}$$

- malha colocalizada Fig.1(a): as variáveis u, v, w e P estão todas no mesmo ponto, ponto Central do domínio.

- malha deslocada Fig.1(b,c,d): as variáveis u, v e w estão todas no defasadas de 1/2 volume em relação ao domínio prinicipal.

A malha referente a velocidade u pode ser deslocada para direita ou esquerda, na direção horizontal. A malha referente a velocidade V pode ser deslocada para cima ou para baixo, na direção vertical. A malha da pressão é mantida na posição central, pois é a referência para os deslocamentos.

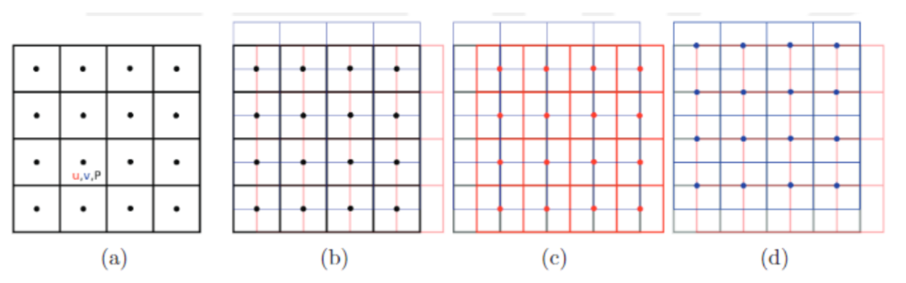

Figure 1: malhas

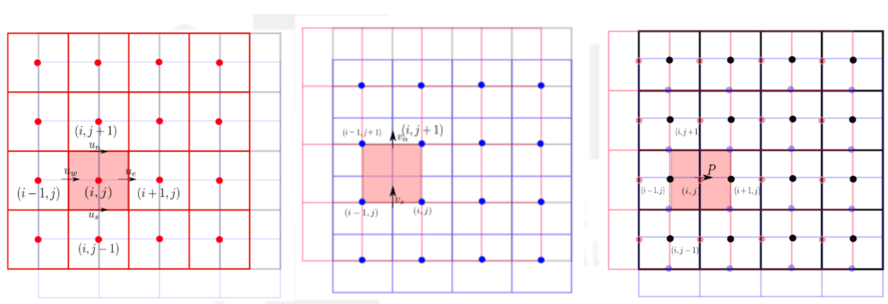

Figure 2: malha deslocada u

Atenção para o volume em destaque, no caso o volume vermelho:

$$\frac{u^*_{i,j} - u^0_{i,j}}{\Delta t} + \tag{8}$$

transiente

$$+\frac{u^0_e u^0_e - u^0_w u^0_w}{\Delta x} = \frac{1}{\Delta x}\left[\left(\frac{u_{i+1,j} + u_{i,j}}{2}\right)^2 - \left(\frac{u_{i,j} + u_{i-1,j}}{2}\right)^2\right] \tag{9}$$

convectivo (parte 1)

$$\frac{u^0_n v^0_n - u^0_s v^0_s}{\Delta y} = \frac{1}{\Delta y}\left[\frac{u_{i,j-1} + u_{i,j}}{2} . \frac{v_{i,j+1} + v_{i-1,j+1}}{2} - \frac{u_{i,j} + u_{i,j-1}}{2} . \frac{v_{i,j} + v_{i-1,j}}{2}\right] \tag{10}$$

convectivo (parte 2)

$$-\frac{1}{\rho}\frac{P^0_{i+1,j} - P^0_{i,j}}{\Delta x} + \nu\left(\frac{u^0_{i+1,j} - 2u^0_{i,j} + u^0_{i-1,j}}{\Delta x^2} + \frac{u^0_{i,j+1} - 2u^0_{i,j} + u^0_{i,j-1}}{\Delta y^2}\right) \tag{11}$$

(pressão + difusivo)

Discretização para a malha de velocidade vertical (v)

$$\frac{\partial v}{\partial t} + \left(\frac{\partial uv}{\partial x} + \frac{\partial vv}{\partial y} + \frac{\partial wv}{\partial z}\right) = -\frac{1}{\rho}\frac{\partial P}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right) \tag{12}$$

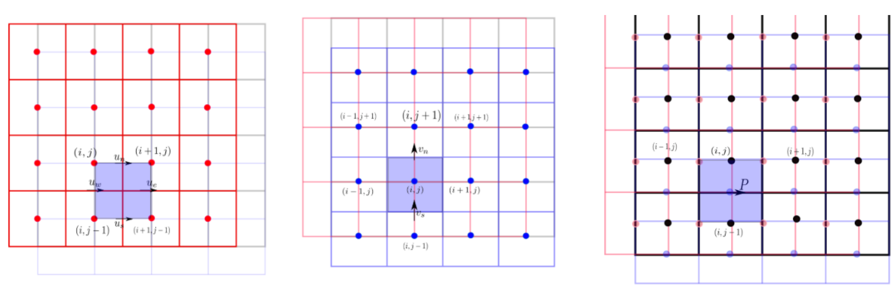

Figure 3: malha deslocada v

$$\frac{v^\star_{i,j} - v^0_{i,j}}{\Delta t} \tag{13}$$

transiente

$$\frac{1}{\Delta x}\cdot\left[\left(\frac{u^0_{(i+1,j)} + u^0_{(i+1,j-1)}}{2}\right)\cdot\left(\frac{v^0_{(i,j)} + v^0_{(i+1,j)}}{2}\right) - \left(\frac{u^0_{(i,j)} + u^0_{(i,j-1)}}{2}\right)\cdot\left(\frac{v^0_{(i,j)} + v^0_{(i-1,j)}}{2}\right)\right] + \tag{14}$$

Convectivo (parte 1)

$$\frac{1}{\Delta y}\cdot\left[\left(\frac{v^0_{(i,j+1)}+v^0_{(i,j)}}{2}\right)^2-\left(\frac{v^0_{(i,j-1)}+v^0_{(i,j)}}{2}\right)^2\right]= \tag{15}$$

Convectivo (parte 2)

$$-\frac{1}{\rho}\cdot\left(\frac{P^0_{(i,j)}-P^0_{(i,j-1)}}{\Delta y}\right) \tag{16}$$

Pressão

$$\nu\cdot\left(\frac{v^0_{(i+1,j)}-2\cdot v^0_{(i,j)}+v^0_{(i-1,j)}}{\Delta x^2}+\frac{v^0_{(i,j+1)}-2\cdot v^0_{(i,j)}+v^0_{(i,j-1)}}{\Delta y^2}\right) \tag{17}$$

Difusivo

1.1 Passo Fracionado

aplica o operador ∇ na equação de Navier-Stokes.

$$\frac{u^\star-u^0}{\Delta t}+\left(\frac{\partial uu}{\partial x}+\frac{\partial uv}{\partial y}+\frac{\partial uw}{\partial z}\right)=-\frac{1}{\rho}\frac{\partial P}{\partial x}+\nu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right) \tag{18}$$

$$\frac{v^\star-v^0}{\Delta t}+\left(\frac{\partial uv}{\partial x}+\frac{\partial vv}{\partial y}+\frac{\partial wv}{\partial z}\right)=-\frac{1}{\rho}\frac{\partial P}{\partial y}+\nu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right) \tag{19}$$

———————— continuamos daqui.

$$u^\star=-\left(\frac{\partial uu}{\partial x}+\frac{\partial uv}{\partial y}+\frac{\partial uw}{\partial z}\right)\Delta t-\frac{1}{\rho}\frac{\partial P}{\partial x}\Delta t+\nu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right)\Delta t+u^0 \tag{20}$$

$$v^\star=-\left(\frac{\partial uv}{\partial x}+\frac{\partial vv}{\partial y}+\frac{\partial wv}{\partial z}\right)\Delta t-\frac{1}{\rho}\frac{\partial P}{\partial y}\Delta t+\nu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right)\Delta t+v^0\Delta t \tag{21}$$

Considerando bidimensional na direção x e y. Portanto w=0

$$u^\star=-\left(\frac{\partial uu}{\partial x}+\frac{\partial uv}{\partial y}\right)\Delta t-\frac{1}{\rho}\frac{\partial P}{\partial x}\Delta t+\nu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)\Delta t+u^0 \tag{22}$$

$$v^\star = -\left(\frac{\partial uv}{\partial x} + \frac{\partial vv}{\partial y}\right)\Delta t - \frac{1}{\rho}\frac{\partial P}{\partial y}\Delta t + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)\Delta t + v^0 \tag{23}$$

Equação de Poisson
Considernado

$$F = -\left(\frac{\partial uu}{\partial x} + \frac{\partial uv}{\partial y}\right)\Delta t + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)\Delta t \tag{24}$$

$$G = -\left(\frac{\partial uv}{\partial x} + \frac{\partial vv}{\partial y}\right)\Delta t + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)\Delta t \tag{25}$$

$$\frac{u^t - u^0}{\Delta t} = -\frac{1}{\rho}\frac{\partial P^t}{\partial x}\Delta t + F^0 \tag{26}$$

$$\frac{u^\star - u^0}{\Delta t} = -\frac{1}{\rho}\frac{\partial P^0}{\partial x}\Delta t + F^0 \tag{27}$$

subtraindo Eq. 26 com Eq. 27

$$\frac{u^t - u^0}{\Delta t} - \frac{u^\star - u^0}{\Delta t} = -\frac{1}{\rho}\frac{\partial P^t}{\partial x}\Delta t + F^0 - \left(-\frac{1}{\rho}\frac{\partial P^0}{\partial x}\Delta t + F^0\right) \tag{28}$$

simplificando

$$\frac{u^t - u^\star}{\Delta t} = -\frac{1}{\rho}\frac{\partial P^t}{\partial x}\Delta t + \frac{\partial P^0}{\partial x}\Delta t \tag{29}$$

$$\frac{u^t - u^\star}{\Delta t} = -\frac{\Delta t}{\rho}\left[\frac{\partial P^t}{\partial x} + \frac{\partial P^0}{\partial x}\right] \tag{30}$$

Sabendo que $P^\star = P^t + P^0$, aplicando o operador gradiente,

$$\frac{\partial P^\star}{\partial x} = \frac{\partial P^t}{\partial x} + \frac{\partial P^0}{\partial x} \tag{31}$$

Substituindo Eq. 31 em Eq. 30

$$u^t - u^\star = -\frac{\Delta t}{\rho}\left[\frac{\partial P^\star}{\partial x}\right] \tag{32}$$

fazendo a mesma coisa para a direção vertical e somando as equações

$$u^t - u^\star + v^t - v^\star = -\frac{\Delta t}{\rho}\left[\frac{\partial P^\star}{\partial x} + \frac{\partial P^\star}{\partial y}\right] \tag{33}$$

aplicando o operador divergente

$$\frac{\partial u^t}{\partial x} - \frac{\partial u^\star}{\partial x} + \frac{\partial v^t}{\partial y} - \frac{\partial v^\star}{\partial y} = -\frac{\Delta t}{\rho}\left[\frac{\partial^2 P^\star}{\partial x^2} + \frac{\partial^2 P^\star}{\partial y^2}\right] \tag{34}$$

aplicando a equação da conservação da massa $\frac{\partial u^t}{\partial x} + \frac{\partial v^t}{\partial y} = 0$.

$$-\frac{\partial u^\star}{\partial x} - \frac{\partial v^\star}{\partial y} = -\frac{\Delta t}{\rho}\left[\frac{\partial^2 P^\star}{\partial x^2} + \frac{\partial^2 P^\star}{\partial y^2}\right] \tag{35}$$

reorganizando

$$\frac{\partial u^\star}{\partial x} + \frac{\partial v^\star}{\partial y} = \frac{\Delta t}{\rho}\left[\frac{\partial^2 P^\star}{\partial x^2} + \frac{\partial^2 P^\star}{\partial y^2}\right] \tag{36}$$

Correção dos campos de velocidade e pressão

$$P^t = P^0 + P^\star \tag{37}$$

$$u^t = u^\star - \frac{\Delta t}{\rho}\left[\frac{\partial P^\star}{\partial x}\right] \tag{38}$$

$$v^t = v^\star - \frac{\Delta t}{\rho}\left[\frac{\partial P^\star}{\partial y}\right] \tag{39}$$

Discretizando 36:

1.2 resumo

- estima campo de velocidade $u^\star$ e $v^\star$
- obtenha o campo de pressão estimado $P^\star$
- atualize o campo de velocidade, u^t e v^t
- atualize o campod pressão $P^t = P^\star + P^0$
- checar a conservação da massa
- avançar no tempo, se o item anterior foi atendido.

———————— continuamos daqui.
Abrir a equação de Poisson utilizando as diferenças finitas centradas

$$\frac{u^\star{}_{(i+1,j)} - u^\star{}_{(i,j)}}{\Delta x} + \frac{v^\star{}_{i,j+1} - v^\star{}_{(i,j)}}{\Delta y} = \frac{\Delta t}{\rho}\left[\frac{P^\star{}_{(i+1,j)} - 2 * P^\star_{(i,j)} + P^\star_{(i-1,j)}}{\Delta x^2} + \frac{P^*_{(i,j+1)} - 2 * P^\star{}_{(i,j)} + P^\star{}_{(i,j-1)}}{\Delta y^2}\right] \tag{40}$$

montar a matriz coeficientes A.